



Lecture 9: Rocky Planets I

Earth & Venus

Planetary Systems

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Learning Objectives

By the end of this lecture, you will be able to:

- Use Earth's interior, surface, ocean and climate as the reference case
- Compare Venus to Earth in mass, radius, atmosphere and rotation
- Explain why a saturated water atmosphere has a maximum thermal flux
- Derive the Simpson-Nakajima runaway greenhouse limit
- Read the D/H ratio as evidence for past water on Venus



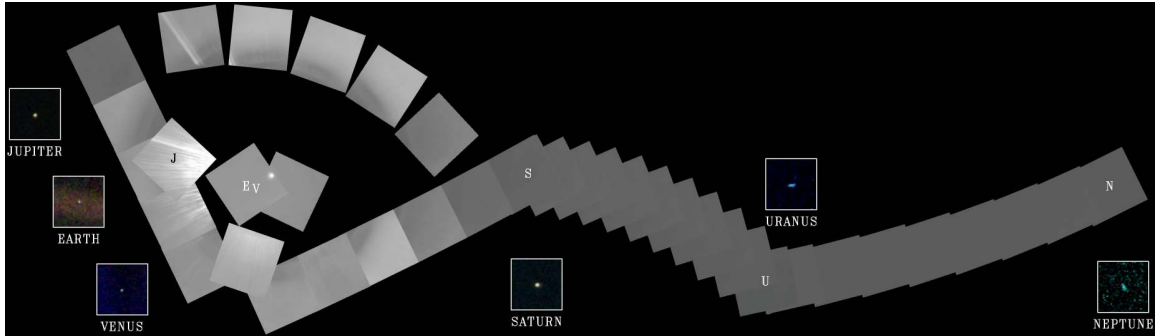
Mercury, Venus, Earth, Mars relative scale. Inner rocky planets sharing similar accretion zones but diverging into radically different surface environments. Credit: NASA/JPL.

Recap: Lectures 1–8

The two innermost large rocky planets share similar bulk parameters but followed radically divergent evolutionary paths.

- Differentiation separates crust, mantle, and core ($C/MR^2 = 0.331$ for Earth)
- Volcanism, impact cratering, and escape shape surfaces and atmospheres
- Liquid outer core sustains a protective magnetic dynamo

Today: What caused the evolutionary divergence of Earth and Venus?



Voyager 1 Solar System family portrait (1990). Inset frames show the planets detected across the outer and inner Solar System, illustrating planetary divergence from a shared disk. Credit: NASA/JPL-Caltech.

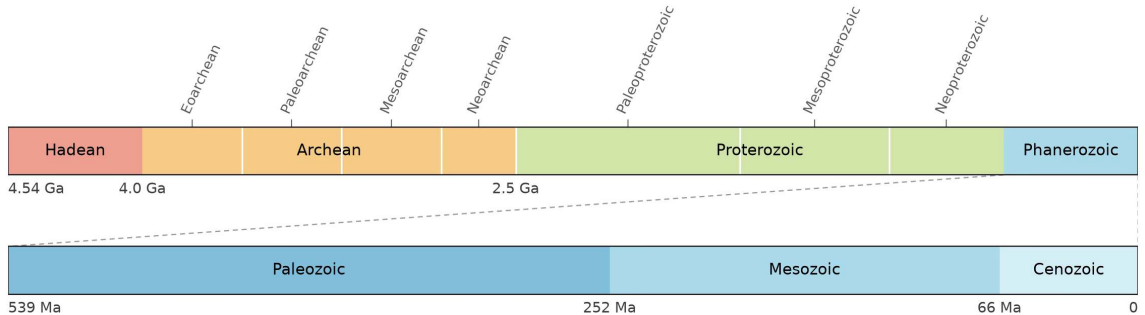
1. Earth as Reference

Earth seen from ~ 6 billion km. Voyager 1 (1990). Credit: NASA/JPL-Caltech

Earth at a Glance

Parameter	Value	Note
Mass	$5.972 \times 10^{24} \text{ kg}$	defines $M_{\oplus}$
Radius (mean)	6371 km	defines $R_{\oplus}$
Mean density	5515 kg m^{-3}	uncompressed $\sim 4030 \text{ kg m}^{-3}$
Semimajor axis	1.000 AU	defines AU
Bond albedo	0.30	cloud + ice + land
T_{eq}	255 K	equilibrium, no greenhouse
T_{surf}	288 K	+33 K greenhouse warming
Rotation period	23.93 h	spin axis tilt 23.4°
Magnetic field	dipolar, $\sim 25\text{--}65 \mu\text{T}$	active dynamo

Source: NASA Earth Fact Sheet (NSSDC)



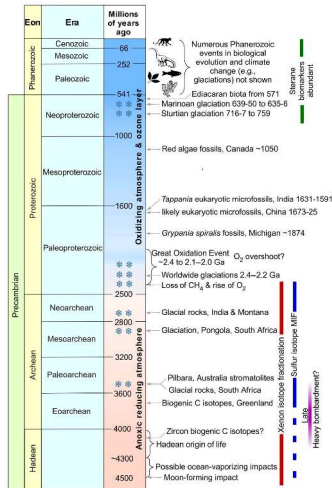
Geologic eons and eras of Earth across 4.54 Gyr of planetary history. The informal Precambrian occupies almost 90% of the timeline before the Phanerozoic diversification of macroscopic life. Credit: Gradstein et al. (2020).

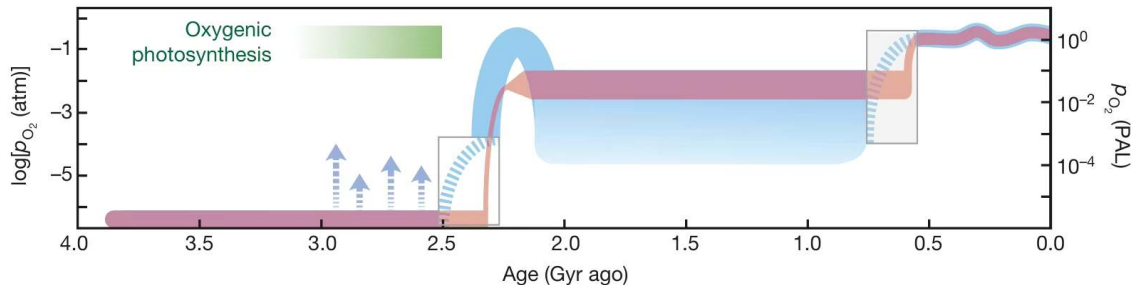
What Makes Earth Habitable?

Four ingredients, all present and stable for ~ 4 Gyr:

1. **Liquid water** on the surface, continuously
2. **Plate tectonics**: long-term carbon cycle, climate buffering
3. **Magnetic field**: shields atmosphere from solar wind erosion
4. **Stable orbit**: low eccentricity, large stabilising Moon

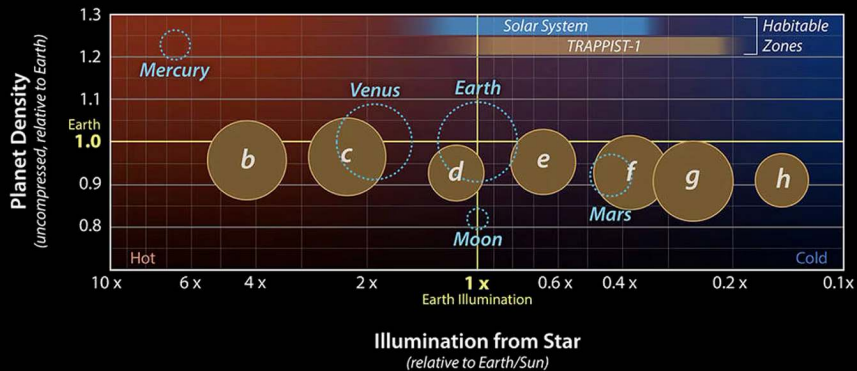
Identify which planetary features are necessary, which are sufficient, and which are accidental.





Atmospheric oxygen through Earth history on a logarithmic scale relative to the present level: essentially anoxic (below 10^{-5} of present) for the first about 2 Gyr, an abrupt rise at the Great Oxidation Event near 2.4 Ga, a plateau at intermediate values through the mid-Proterozoic, and a second rise to near-modern levels in the Neoproterozoic and Phanerozoic, tracking the growth of the photosynthetic biosphere and the redox balance with the deep ocean and crust. Reproduced from Lyons et al. (2014).

TRAPPIST-1/Solar System Comparison



Credit: NASA

TRAPPIST-1 vs. inner solar system, to scale. Seven Earth-sized worlds span the habitable zone around a late M dwarf, testing terrestrial evolutionary pathways. Credit: NASA/JPL-Caltech (PIA22093).

Plate Tectonics: The Engine

Earth alone among known solar system bodies operates active plate tectonics today, driving continuous volatile recycling between surface and interior.

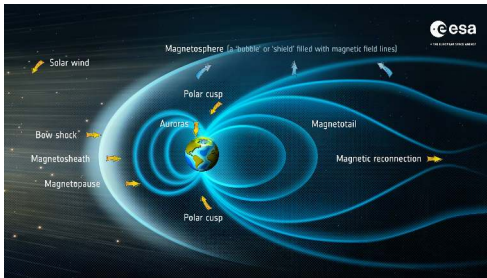
- Lithosphere divided into ~ 15 major plates moving at cm yr^{-1} speeds
- Subduction zones recycle crust and surface volatiles on ~ 100 Myr timescales
- Closes the long-term carbon cycle by returning buried carbonate to the mantle



Global glaciation during Cryogenian Snowball Earth episodes. High ice albedo maintained freezing conditions until volcanic outgassing and carbonate-silicate feedbacks forced deglaciation. Credit: Oleg Kuznetsov, Wikimedia Commons, CC BY-SA 4.0.

Earth's Magnetic Field

A vigorous geodynamo generates a protective dipolar magnetosphere that shields Earth's atmosphere from direct solar wind erosion.



Credit: ESA, CC BY-SA 3.0 IGO

- Surface dipolar intensity of $25\text{--}65\ \mu\text{T}$ driven by core convection
- Magnetopause dayside stand-off distance at $\sim 10 R_{\oplus}$
- Deflects solar wind and suppresses ion-pickup escape of volatiles

Hydrosphere and Cryosphere

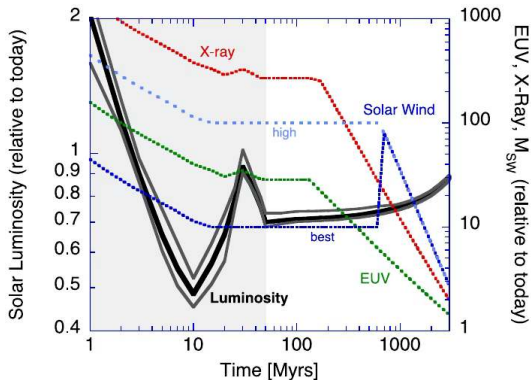


Credit: NASA / Apollo 17 crew

Earth possesses a stable surface liquid ocean containing over a billion cubic kilometres of water, complemented by polar cryosphere reservoirs.

- Oceans contain $1.34 \times 10^9 \text{ km}^3$ of liquid water
- Total surface water equivalent to $\approx 2700 \text{ m}$ global equivalent layer (GEL)
- Deep mantle holds $\geq 1\text{--}2$ additional ocean masses

Climate Stability and Solar Evolution

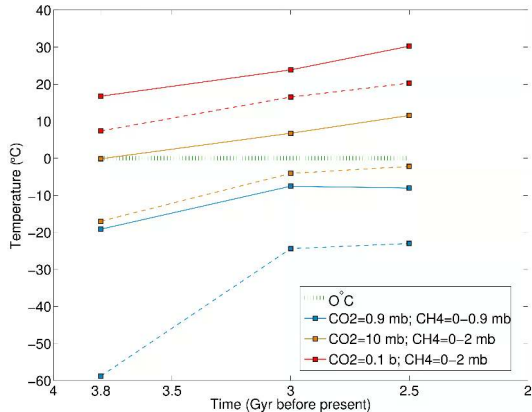


Credit: Zahnle et al. (2007)

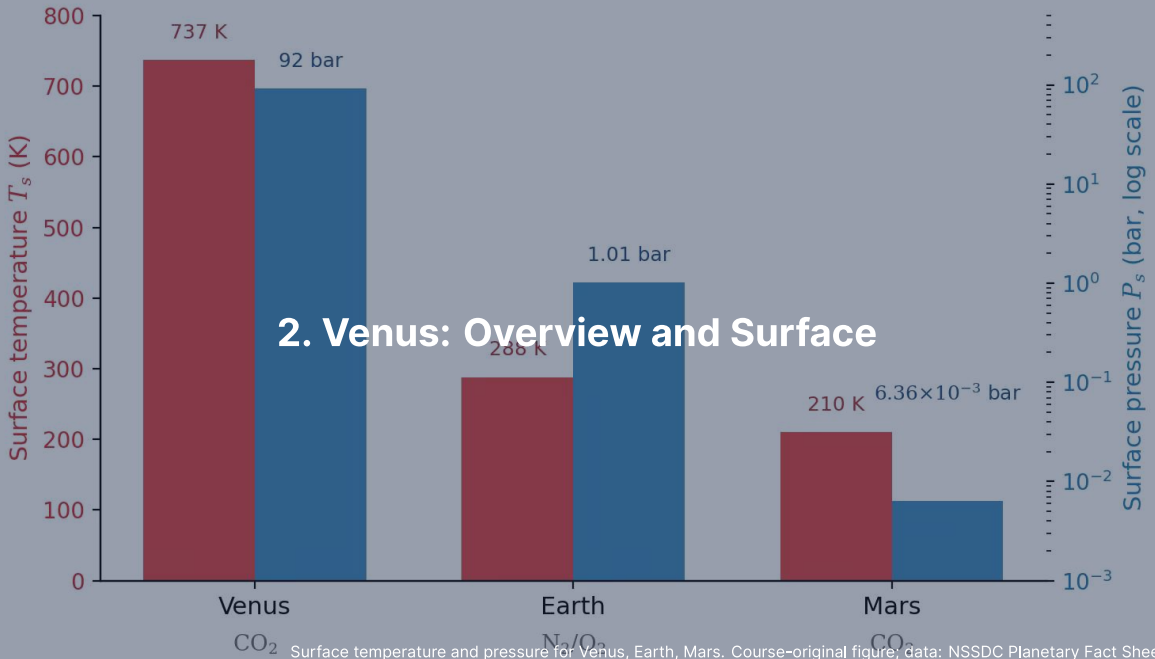
Faint young Sun problem:

- Solar luminosity increased by ~ 30% since 4.5 Ga
- Liquid water persisted at ≥ 3.8 Ga despite dimmer early Sun
- Carbonate-silicate negative feedback stabilized climate

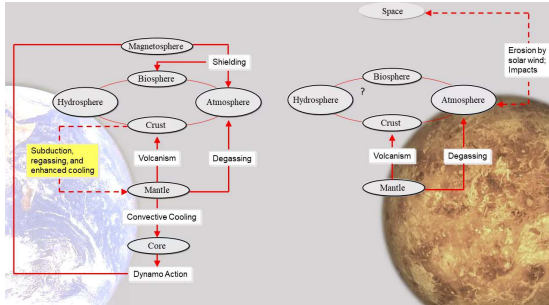
Habitable zone inner edge migrated outward over time, sweeping past Venus.



Global mean surface temperature of the Archean Earth from 3.8 to 2.5 Ga in a 3D general-circulation model for three compositions: 0.9 mbar CO₂ with 0.9 mbar CH₄ (blue), 10 mbar CO₂ with 2 mbar CH₄ (orange), 0.1 bar CO₂ with 2 mbar CH₄ (red); solid with methane, dashed without; the dotted green line is the freezing point. A few mbar of CO₂ is marginal at 3.8 Ga; about 0.1 bar resolves the faint-young-Sun problem at all three epochs. Reproduced from Charnay et al. (2013).



Earth and Venus as Coupled Systems



Credit: Gillmann et al. (2022)

Both are coupled systems of:

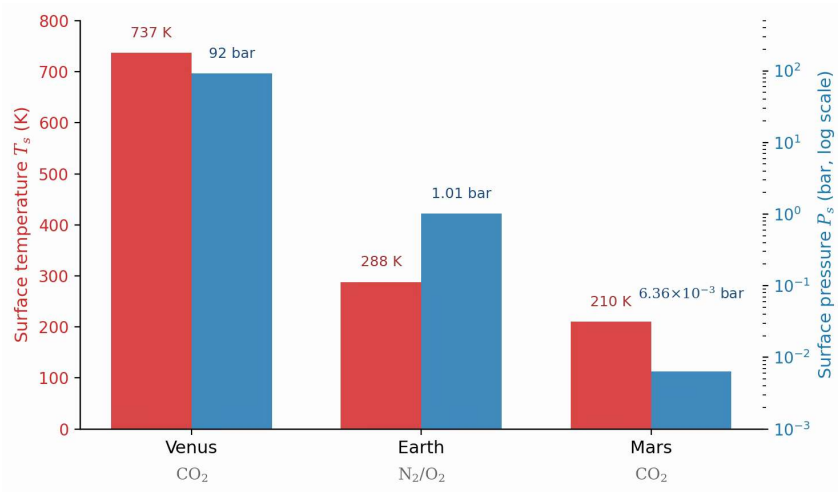
- Interior + tectonics
- Magnetosphere or its absence
- Atmosphere + clouds
- Surface + (possible) hydrosphere

The differences between them are emergent properties of which feedback loops close.

Venus and Earth: Twin and Not

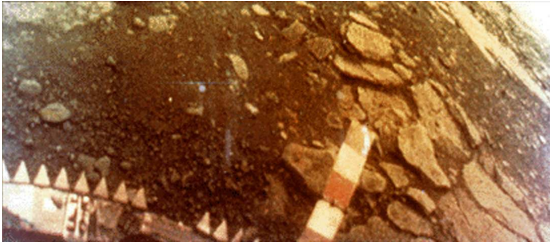
Parameter	Earth	Venus
Mass ($M_{\oplus}$)	1.000	0.815
Radius ($R_{\oplus}$)	1.000	0.949
Mean density (kg m^{-3})	5515	5243
Semimajor axis (AU)	1.000	0.723
Stellar flux ($S_{\oplus}$)	1.00	1.91
Bond albedo	0.30	0.77
T_{surf} (K)	288	737
Surface pressure (bar)	1.0	92
Atmosphere	N_2/O_2	CO_2/N_2
Rotation period (d)	+1.0	-243 (retrograde)
Magnetic field	$\sim 50 \mu\text{T}$	$< 10^{-5} \times \text{Earth}$

Source: NASA Venus Fact Sheet (NSSDC)



Surface temperature and surface pressure comparison for Venus, Earth, and Mars. Near-twins in bulk size and density, Venus and Earth differ by a factor of 92 in pressure and 450 K at the surface. Credit: Course-original figure; data from NASA Fact Sheet.

Venus: A Long Mission History

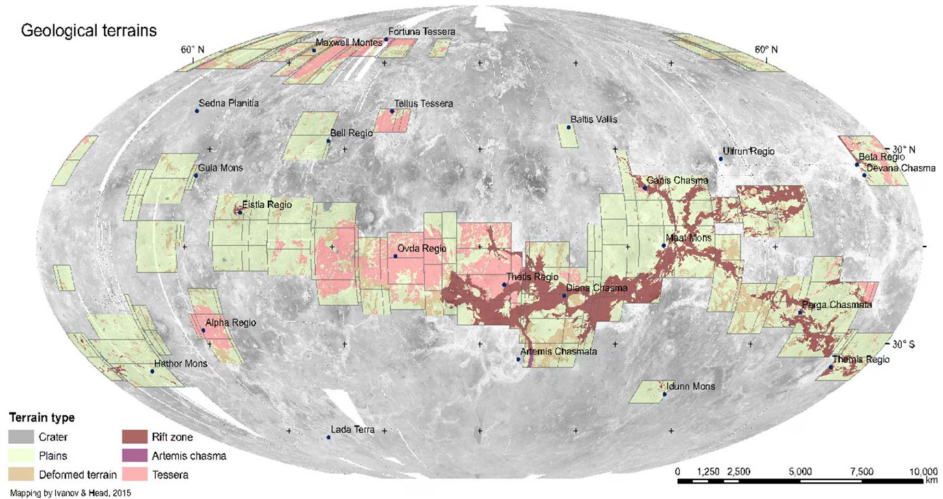


Credit: USSR / NASA NSSDC, public domain

Hostile surface conditions of 92 bar and 737 K limited lander lifespans to two hours, focusing exploration on radar and orbital remote sensing.

- **Venera** (1975–1982): first surface images and XRF basaltic composition
- **Magellan** (1990–1994): global radar mapping at ~ 100 m resolution
- **Venus Express & Akatsuki** (2006 to present): atmospheric dynamics, escape rates (Venus Express) and super-rotation

Geological terrains

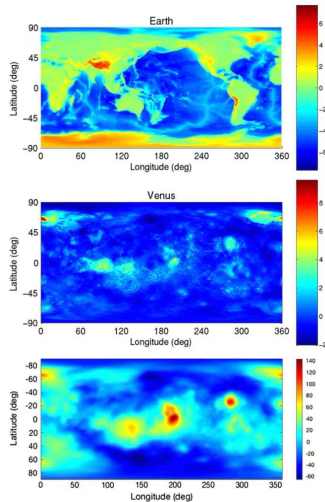


Magellan radar mosaic with terrain overlay. Volcanic plains cover ~80% of the surface alongside tessera highlands, showing no plate-tectonic boundaries. Widemann et al. (2023), SSRv 219, 56.

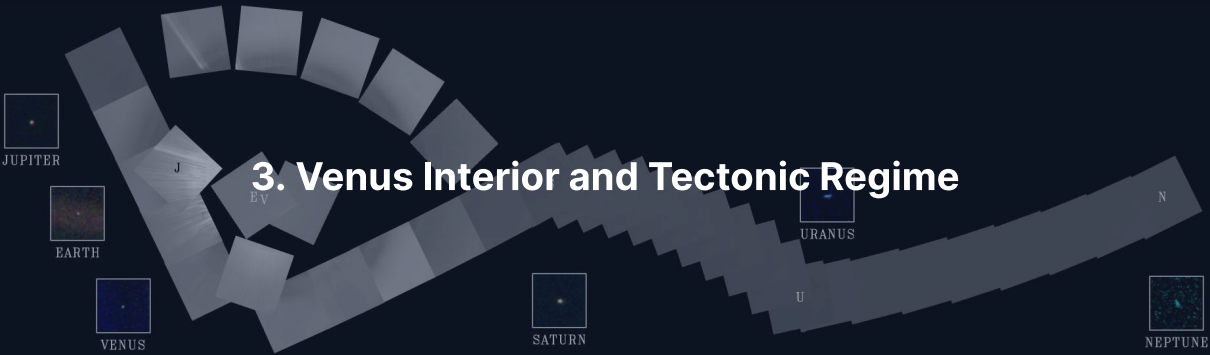
Crater Statistics: A Single Surface Age

A statistically random distribution of ~ 1000 craters yields a young global crater-retention age of 150–250 Myr across all terrains.

- ~ 1000 catalogued impact craters (≥ 2 km) show no spatial clustering
- Uniform density implies a young surface age of ~ 150–250 Myr
- Debated between catastrophic global resurfacing and steady volcanic equilibrium



Topography and geoid comparison. Unimodal hypsometry on Venus contrasts with Earth's bimodal ocean-continent distribution, ruling out Earth-style plate tectonics. Smrekar et al. (2018).



Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Venus' Interior: Inferred, Not Measured

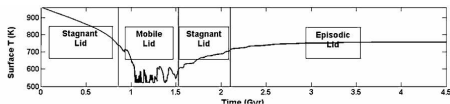
Without landed seismology or precession measurements, Venus' internal structure is modeled by Earth analogy rather than directly measured.

- Moment of inertia factor $C/MR^2 \approx 0.33$ is assumed, not measured
- Inferred metallic core radius $R_c \approx 3000\text{--}3500\text{ km}$ ($\sim 0.55 R_V$)
- Mean density of 5243 kg m^{-3} requires a substantial iron core

Venus Tectonic Evolution: A Model Trajectory

Surface temperature evolution from one numerical model: successive intervals of

- **Stagnant lid**
- **Mobile lid**
- **Stagnant lid**
- **Episodic lid**



Credit: Smrekar et al. (2018)

Conceptual alternatives debated for Venus today: **steady stagnant** (localised mantle plumes) vs. **episodic catastrophic overturn**. Either way, no Earth-style plate tectonics.

The Missing Magnetic Field

Venus lacks an intrinsic dipole magnetic field, leaving its upper atmosphere exposed to direct solar wind interaction and erosion.

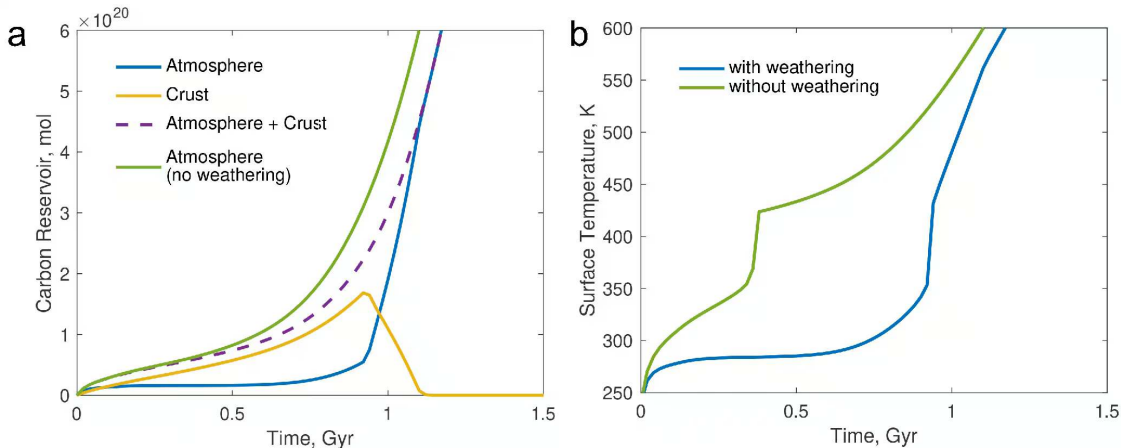
- Internal magnetic field measured at $< 10^{-5}$ times Earth's field
- Stagnant-lid insulation suppresses core heat loss, shutting down thermal convection
- Unshielded atmosphere enables solar-wind induced escape and ion pickup

Why Stagnant Lid?

A hot, dry surface produces a strong lithosphere that resists subduction, locking Venus into an immobile stagnant-lid convective regime.

- Plate subduction requires a hydrated, mechanically weakened lithosphere
- High surface temperatures (737 K) promote ductile flow rather than brittle faulting
- No subduction return leg breaks the long-term carbon cycle permanently

Source: Smrekar et al. (2018); Gillmann et al. (2022)



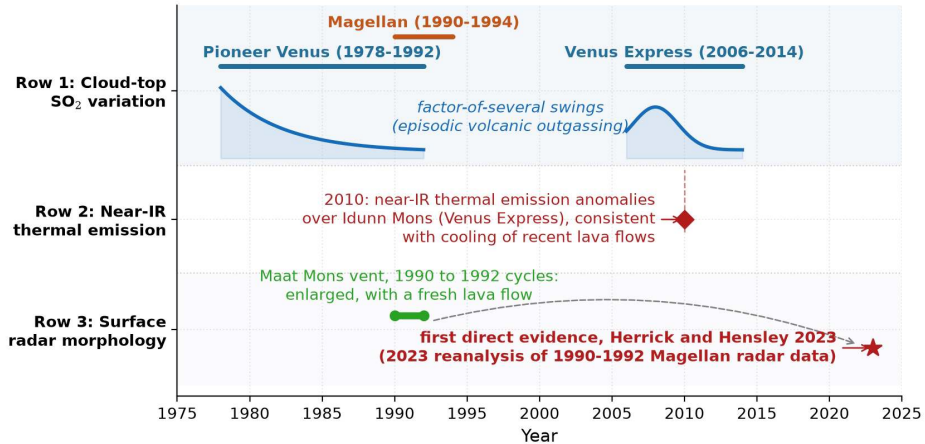
Coupled interior-atmosphere evolution of a stagnant-lid Venus. Surface weathering sustains temperate conditions for 0.9 Gyr until water loss triggers runaway crustal decarbonation to a 92-bar end state. Credit: Honing et al. (2021).

Is Venus Volcanically Active Today?

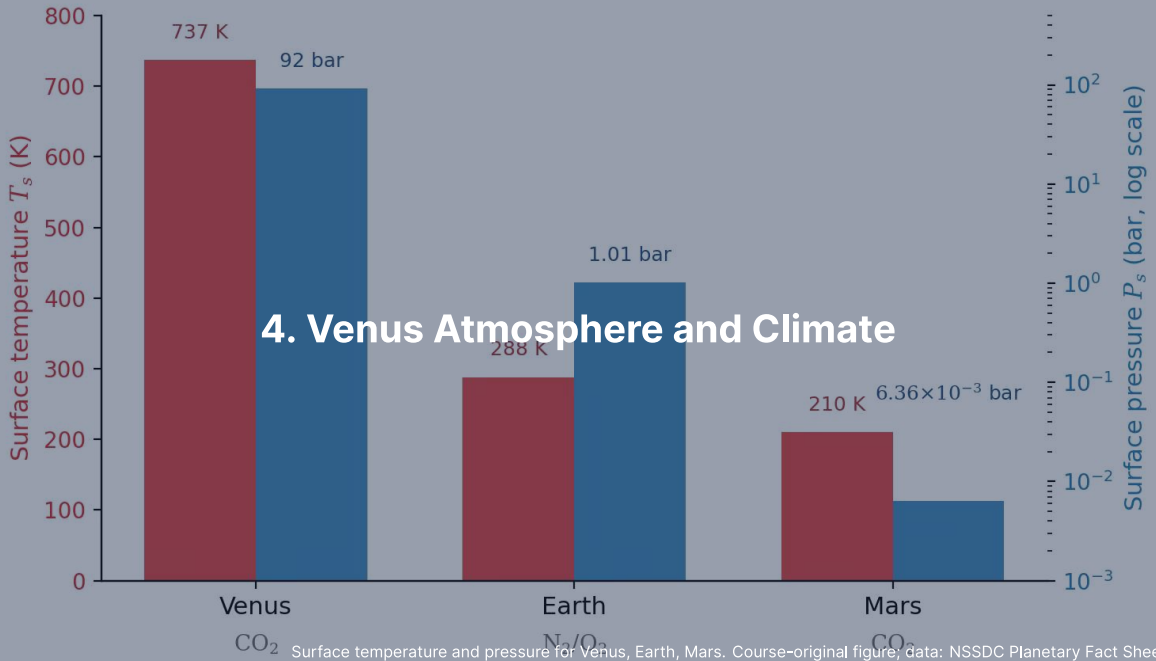
Radar and orbital observations demonstrate that Venus remains volcanically active today, driven by internal heat loss through mantle plumes.

- Magellan radar reanalysis reveals a $\sim 2 \text{ km}^2$ vent change on Maat Mons (1991–1992)
- Venus Express thermal emissivity anomalies indicate lava flows younger than 250 kyr
- Decadal fluctuations in atmospheric SO_2 suggest episodic volcanic outgassing

Evidence for present-day volcanic activity on Venus (schematic timeline)



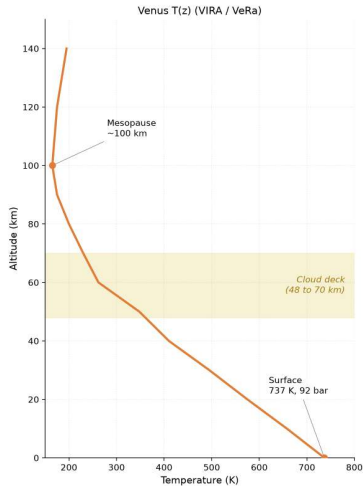
Three lines of evidence for present-day volcanism on Venus on one timeline: cloud-top SO₂ swings, Venus Express thermal anomalies at Idunn Mons, and the Maat Mons vent change in Magellan radar (Herrick and Hensley 2023). Course figure.



Venus Atmosphere: Structure

Venus hosts a massive 92-bar carbon dioxide atmosphere enveloped in global sulfuric acid clouds without a tropopause cold trap.

- Composition: 96.5% CO₂, 3.5% N₂ at 737 K and 92 bar surface pressure
- Global cloud deck at 48–70 km composed of ~ 75% sulfuric acid droplets
- Lower atmosphere follows a dry adiabat with lapse rate $\Gamma \approx 8 \text{ K km}^{-1}$; scale height $H \approx 15.9 \text{ km}$
- $^{40}\text{Ar}/^{36}\text{Ar} \approx 1.1$ (Earth ~ 300): Venus has degassed only a fraction of its interior



Venus atmospheric temperature profile from the 737 K, 92 bar surface through the cloud deck (48–70 km) to 100 km, showing a monotonic dry adiabatic lapse rate. Credit: Pioneer Venus / VIRA and Venus Express; course-original figure.

Atmospheric Super-Rotation

Venus exhibits atmospheric super-rotation, where high-altitude cloud winds encircle the planet sixty times faster than the underlying solid surface.

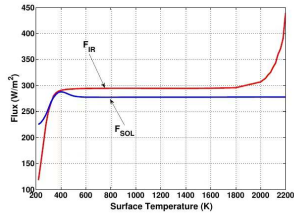
- Cloud-top winds of $\sim 100 \text{ m s}^{-1}$ circle the planet in ~ 4 days
- Solid-body retrograde rotation takes 243 days (60× angular velocity ratio)
- Maintained by wave transport and atmospheric thermal tides

Venus' Greenhouse: T_{surf} from T_{eq}

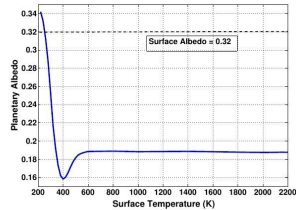
Despite absorbing less solar energy than Earth due to high albedo, Venus experiences extreme greenhouse warming from its massive carbon dioxide atmosphere.

$$T_{\text{eq}} = \left[\frac{S(1 - A)}{4\sigma} \right]^{1/4} \approx 227 \text{ K}, \quad T_{\text{surf}} = 737 \text{ K}$$

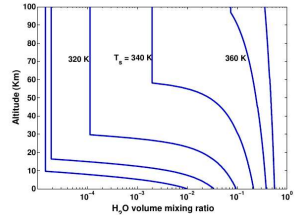
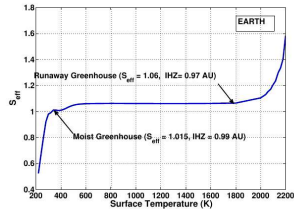
- Bond albedo $A = 0.77$ reflects most sunlight, yielding $T_{\text{eq}} = 227 \text{ K}$
- One-layer greenhouse $T_s = 2^{1/4} T_{\text{eq}} \approx 270 \text{ K}$ falls far short
- Greenhouse warming $\Delta T = +510 \text{ K}$ dwarfs Earth's $+33 \text{ K}$
- Driven by 92 bar of CO_2 , with SO_2 , H_2O and the clouds closing the spectral windows



(a)

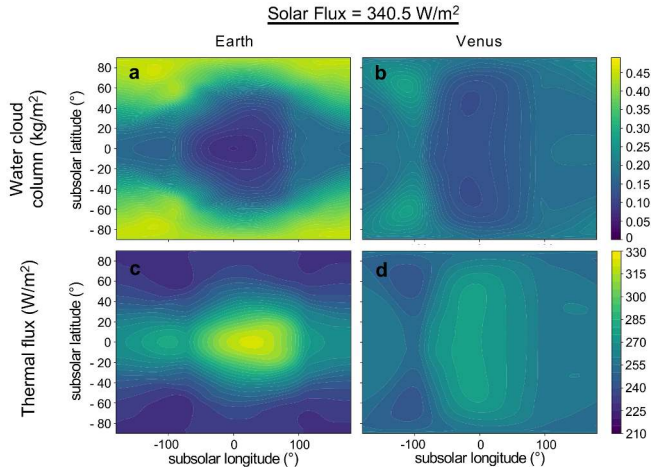


(b)



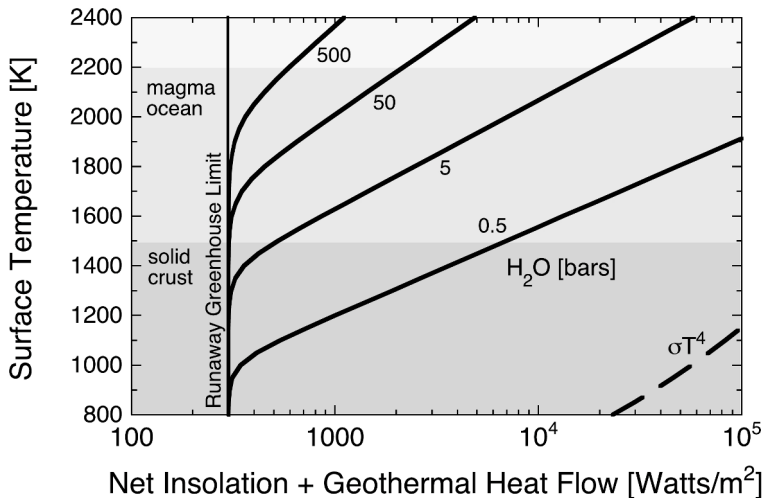
Climate-model calculation of the runaway greenhouse threshold. Outgoing longwave radiation asymptotes to 282–291 W/m^2 as water vapour closes infrared spectral windows at high surface temperature. Credit: Kopparapu et al. (2013).

Where the Night-Side Clouds Form



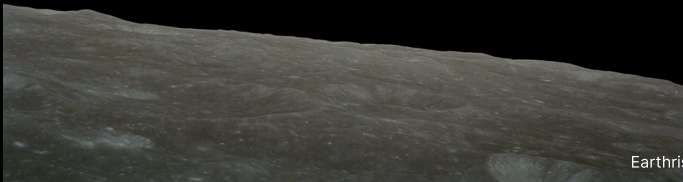
- Hot and steamy Earth (a, c) and Venus (b, d) at the same flux, 340.5 W/m^2
- Clouds gather on the night-side in both, although Venus turns 243 times slower: the pattern does not need slow rotation
- Night-side clouds trap thermal radiation, so the emission maps are anticorrelated with the cloud maps

Credit: Turbet et al. (2021), Fig. 2a-d



Thermal radiation limit and runaway greenhouse threshold. A saturated atmosphere has a maximum OLR of $\sim 280\text{--}300 \text{ W m}^{-2}$, beyond which surface water vaporises completely. Zahnle et al. (2007).

Break



5. Blackboard Derivation: Simpson-Nakajima Limit



Voyager 1 Solar System family portrait (1990); each of the six planets it detected appears in a small inset box. Credit: NASA/JPL-Caltech

Setup: Saturated Atmosphere, Greybody

Goal: show that a water-saturated atmosphere has a maximum OLR independent of surface temperature.

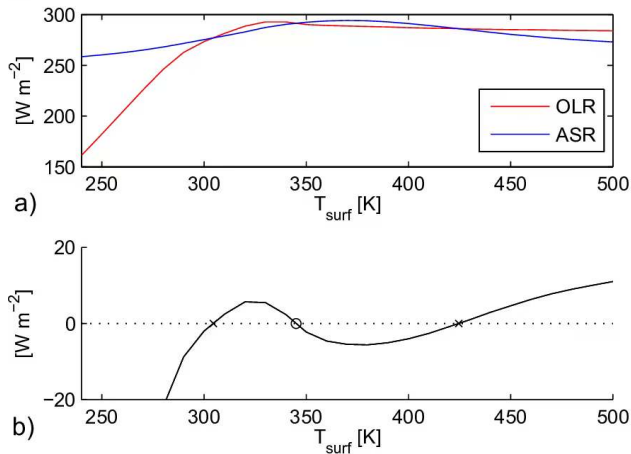
Assumptions:

- Atmosphere saturated in water vapour everywhere
- Single grey IR opacity κ , units $\text{m}^2 \text{kg}^{-1}$
- Hydrostatic equilibrium, surface gravity g
- Photosphere: where IR optical depth $\tau \sim 1$

Clausius-Clapeyron sets the water vapour partial pressure:

$$\frac{dp_{\text{sat}}}{dT} = \frac{L p_{\text{sat}}}{R_v T^2}$$

with $L \approx 2.5 \times 10^6 \text{ J kg}^{-1}$, $R_v = 461 \text{ J kg}^{-1} \text{ K}^{-1}$.



Outgoing longwave radiation and absorbed shortwave radiation equilibria as a function of surface temperature, showing multiple thermal equilibria in saturated greenhouse atmospheres. Credit: Wordsworth et al. (2013).

Result: The OLR Asymptote

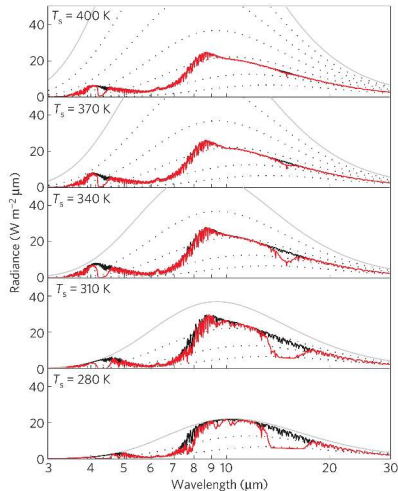
Optical depth unity fixes the photosphere pressure, saturation fixes its temperature (board):

$$p_{\text{phot}} = \frac{g}{\kappa} = p_{\text{sat}}(T_{\text{phot}}) \Rightarrow T_{\text{phot}} \approx 260 \text{ K}$$

$$F_{\text{OLR}}^{\text{max}} \approx \sigma T_{\text{phot}}^4 \approx 5.67 \times 10^{-8} \times (260)^4 \approx 260 \text{ W m}^{-2}$$

Refined line-by-line calculations: $F_{\text{OLR}}^{\text{max}} \approx 280\text{--}310 \text{ W m}^{-2}$ (Kasting 1988; Goldblatt et al. 2013).

As T_s rises, the photosphere migrates upward but its temperature stays clamped: thermal flux saturates.



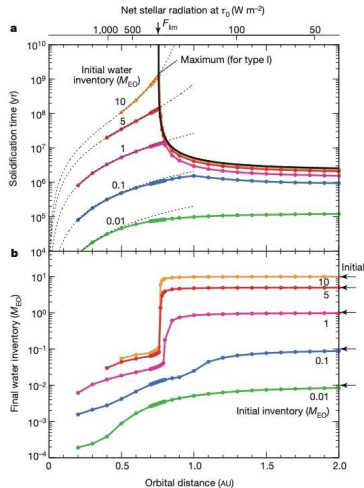
Thermal-radiance spectra of a water-rich atmosphere for surface temperatures from 280 to 400 K, showing flux saturation at the runaway-greenhouse asymptote of 282 W/m^2 as the 8–14 μm window closes. Credit: Goldblatt et al. (2013).

Against the Limit: Earth Below, Venus Above

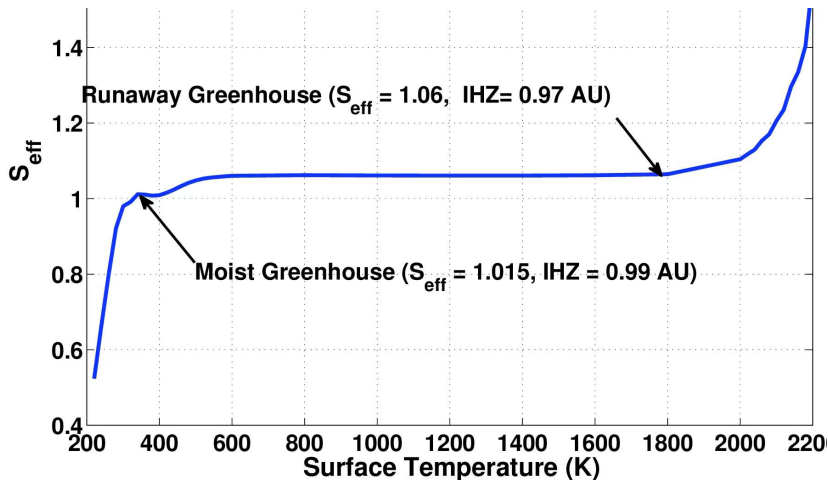
$$F_{\text{abs}} = \frac{S_{\odot}}{4a^2} (1 - A)$$

- Earth ($a = 1$ AU, $A = 0.30$): $F_{\text{abs}} \approx 240 \text{ W m}^{-2}$, below $F_{\text{OLR}}^{\text{max}}$
- Early Venus ($a = 0.723$ AU, Earth-like A): $F_{\text{abs}} \approx 460 \text{ W m}^{-2}$, far above it
- Present Venus ($A = 0.77$) absorbs only $\sim 150 \text{ W m}^{-2}$: the albedo is the product of the runaway, not its cause

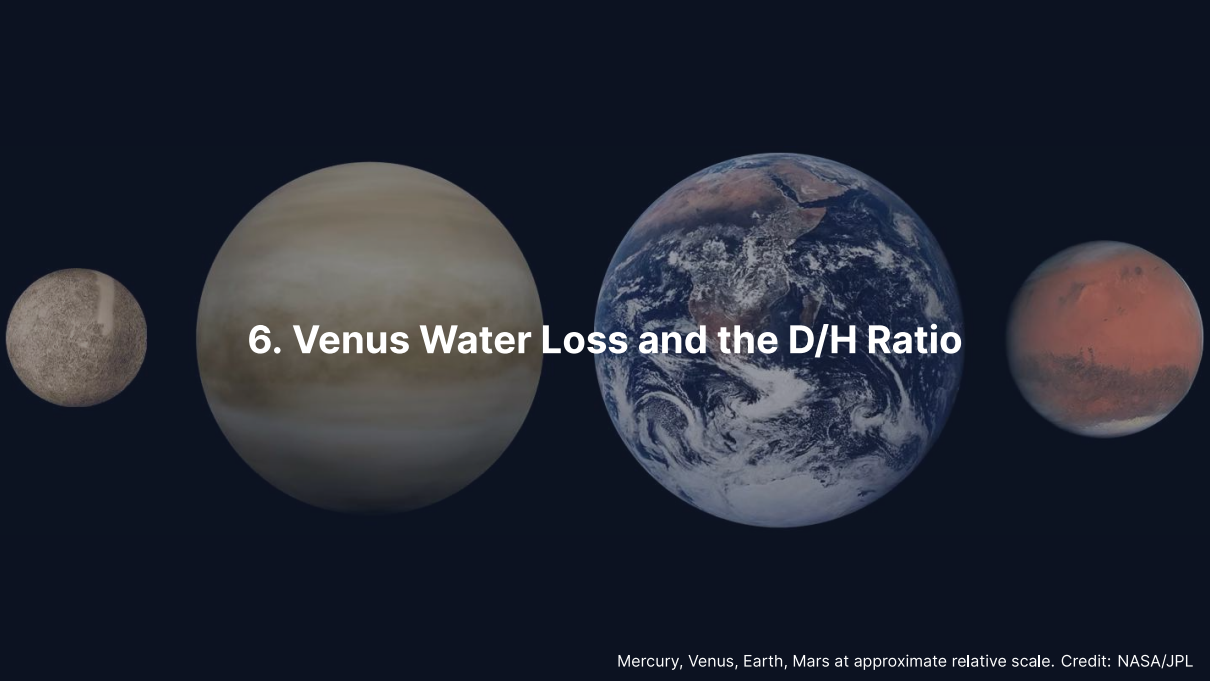
Setting $F_{\text{abs}} = F_{\text{OLR}}^{\text{max}}$ gives the inner edge of the habitable zone,
 $a_{\text{IHZ}} \approx 0.95\text{--}0.99$ AU.



Solidification timescale and final retained water for Type I and Type II terrestrial planets. Beyond the critical runaway distance (0.77 AU), magma oceans solidify rapidly and retain surface oceans. Credit: Hamano et al. (2013).



Inner habitable zone calculation. The runaway greenhouse limits the inner edge to $S_{\text{eff}} = 1.06$ (0.97 AU around the present Sun), placing Venus ($1.91 S_{\oplus}$) well inside. Kopparapu et al. (2013).

A horizontal arrangement of four celestial bodies against a dark blue background. From left to right: Mercury, a small greyish-brown sphere; Venus, a large pale yellowish-tan sphere; Earth, a large blue and white sphere showing continents and clouds; and Mars, a medium-sized reddish-orange sphere. The text '6. Venus Water Loss and the D/H Ratio' is centered over the Venus and Earth spheres.

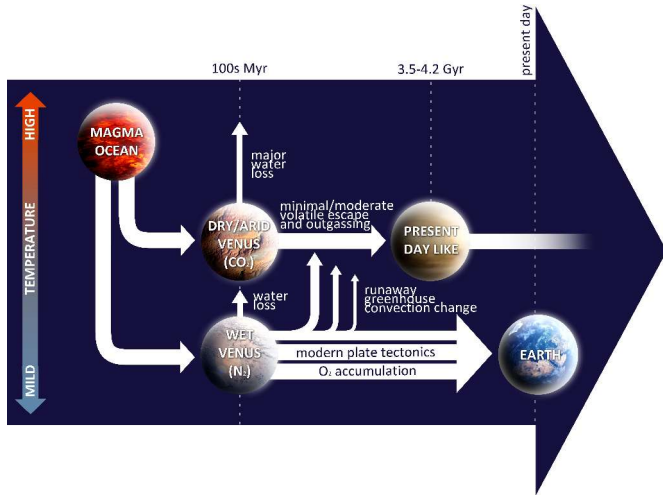
6. Venus Water Loss and the D/H Ratio

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

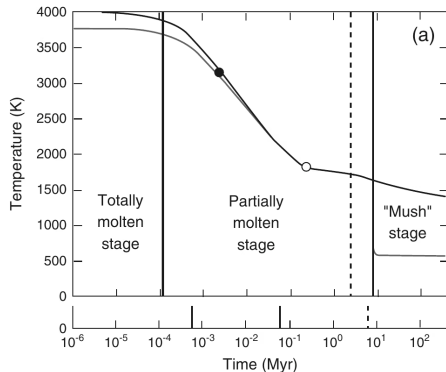
When Did Venus Lose Its Water?

Two hypotheses: early loss from a magma ocean, or late desiccation of a temperate ocean.

- **Early loss (Hamano Type II):** the steam atmosphere never condenses; water escapes during the magma-ocean phase
- **Late loss (Way et al. 2016):** a temperate surface persists for Gyr until solar brightening triggers the runaway
- The timing needs noble-gas isotopes and surface mineralogy from the next missions



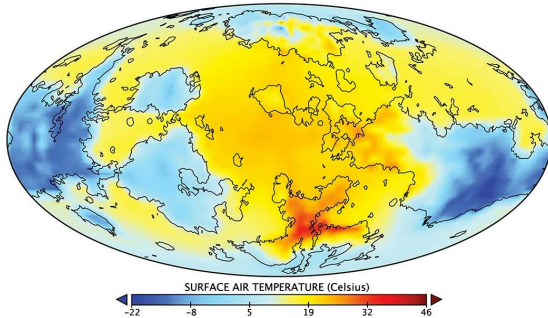
Two end-member evolutionary branches for Venus water history. Early hydrodynamic loss during a magma ocean phase (top branch) contrasts with late desiccation following an extended habitable period (bottom branch). Credit: Gillmann et al. (2022).



Potential temperature (black) and surface temperature (grey) of a crystallising magma ocean coupled to its steam atmosphere on a Venus-mass planet at Venus's orbit; the three regions track the mantle from totally molten to mush, water vapour condenses at the partially-molten to mush boundary, and the dashed line marks 98% solidification at about 10 Myr (1.5 Myr at Earth's orbit, 0.1 Myr at Mars's). The long-lived steam atmosphere has time to photolyse and lose hydrogen: the basis of the early-loss scenario. Below about 0.66 AU the magma ocean never freezes; Venus at 0.72 AU sits just outside that limit.

Was Venus Ever Habitable? The Late-Loss Scenario

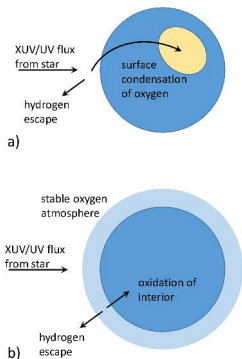
(a) Paleo Venus Surface Air Temperature (2.9Gya solar spectrum)



Credit: Way et al. (2016)

- GCM snapshot at 2.9 Gya: surface T ranges from -22°C polar to $+46^{\circ}\text{C}$ equatorial
- Cloud feedback from slow rotation reflects flux back to space
- Across the Way et al. (2016) ensemble, global mean $T < 290\text{ K}$ can persist for $\sim 1\text{ Gyr}$
- A habitable Venus is dynamically possible, not required

The Loss Mechanism: Photolysis + Escape



Credit: Wordsworth & Pierrehumbert (2014)

Massive water loss proceeds sequentially via stratospheric photolysis, rapid hydrogen escape to space, and residual oxygen oxidation or ion pickup.

- UV photolysis ($\text{H}_2\text{O} + h\nu \rightarrow \text{H} + \text{OH}$) dissociates water in the upper atmosphere
- Light hydrogen escapes efficiently through thermal Jeans and hydrodynamic escape
- Oxygen is absorbed by crustal oxidation or stripped by solar-wind ion pickup

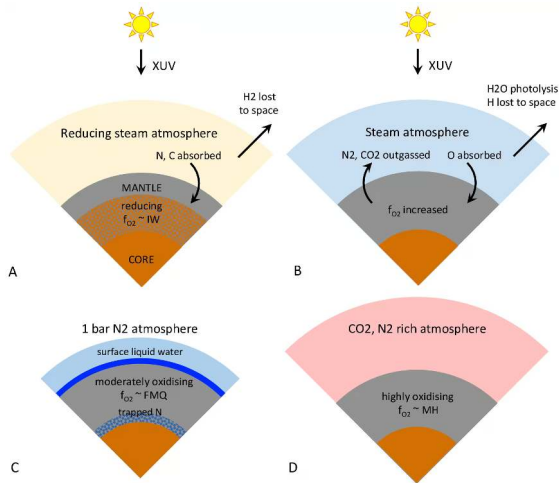
The D/H Ratio: Evidence for Lost Water

Venus' atmospheric D/H is $\sim 150\times$ Earth's: the record of massive past hydrogen escape.

Body	D/H ($\times 10^{-4}$)
Protosolar (H_2)	~ 0.20
Earth (oceans)	1.56
Comets (Halley, 67P)	1.5–5
Venus	~ 240
Mars	~ 8

Source: Donahue et al. (1982); de Bergh et al. (1991)

- Light ^1H escapes more easily than ^2H
- Rayleigh distillation:
 $R/R_0 = f^{\alpha-1}$
- Diffusion-limited ($\alpha = 0.5$):
 $f \approx 150^{-2}$, initial water ~ 450 m GEL; the Jeans limit ($\alpha = 0.71$) gives an unphysical > 100 oceans, so 450 m is a lower bound



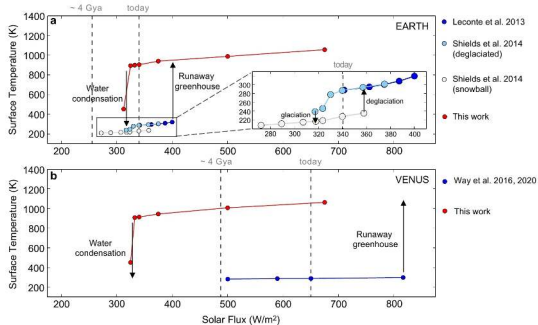
The water-loss redox pump linking atmospheric nitrogen to early water loss. Hydrogen escape drives mantle oxidation and leaves a 92-bar CO₂ atmosphere, contrasting with Earth's 1-bar N₂ state. Credit: Wordsworth (2016).

Bounding the Initial Water Inventory

Rayleigh distillation models translate the measured deuterium enrichment into a firm lower bound on Venus' initial water inventory.

- Rayleigh distillation relation: $R/R_0 = f^{\alpha-1}$ with fractionation factor α
- Diffusion limit ($\alpha \approx 0.5$) yields remaining water fraction
 $f \approx 150^{-2} \approx 4.4 \times 10^{-5}$
- Current 2 cm GEL column implies initial water equivalent layer $d_{\text{init}} \geq 450$ m GEL

Bistability and Hysteresis Loop



Credit: Turbet et al. (2021), Fig. 4

For some stellar fluxes, two stable states coexist:

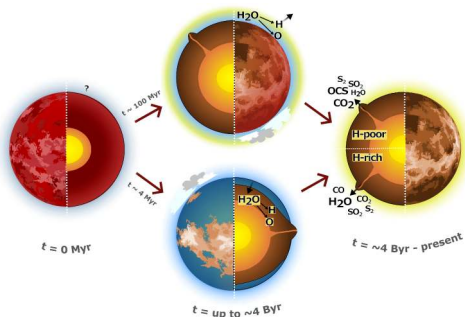
- Wet: ocean + thin atmosphere
- Dry: hot, thick CO_2 atmosphere

Crossing the threshold is one-way. The dry state has no thermodynamic path back.

Constraining the Trajectory

Two end-member pathways for Venus' water history:

- **Top branch (dry):** no surface ocean; H lost early; mantle dry today
- **Bottom branch (wet):** ocean condenses, late loss; mantle still wet today
- Constantinou et al. (2024) argue current photochemistry favours the dry/early-loss branch (volcanic $\text{H}_2\text{O} \lesssim 6\%$)
- DAVINCI noble-gas measurements will discriminate



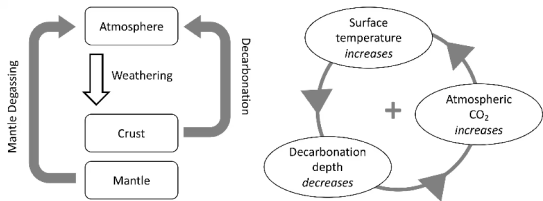
Credit: Constantinou et al. (2024)

The image shows four planets in a horizontal row against a dark blue background. From left to right: Mercury is a small, grey, cratered sphere; Venus is a large, pale yellowish-brown sphere with faint cloud bands; Earth is a large sphere with a blue and white swirling pattern representing oceans and clouds; and Mars is a medium-sized sphere with a reddish-orange surface and some darker patches. The text '7. Why Did Earth and Venus Diverge?' is centered over the Venus and Earth planets in a white, bold, sans-serif font.

7. Why Did Earth and Venus Diverge?

Mercury, Venus, Earth, Mars at approximate relative scale. Credit: NASA/JPL

Carbonate-Silicate Cycle: Failure Mode

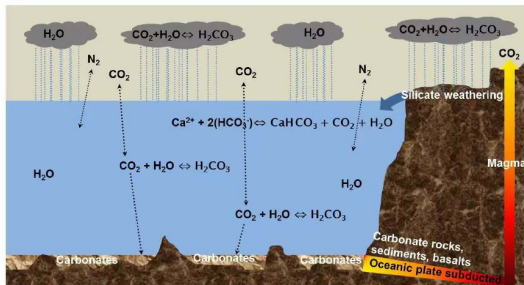


Credit: Honing et al. (2021)

Without surface liquid water and subduction, Venus lacks weathering sinks and mantle return legs, causing volcanic carbon dioxide to build indefinitely.

- Silicate weathering sink collapses entirely in the absence of liquid surface water
- Stagnant-lid regime prevents subduction return of carbonates to the mantle
- Volcanic outgassing irreversibly drives atmospheric CO₂ to the observed 92 bar

The Earth Carbonate-Silicate Schematic



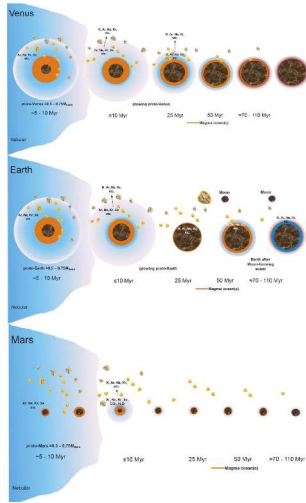
Credit: Lammer et al. (2018)

Earth's closed carbon loop:

- Volcanic outgassing (mantle to atmosphere)
- Silicate weathering on continents (atmosphere to ocean, T -dependent sink)
- Carbonate sedimentation + subduction (ocean to mantle return leg)

Venus today has no liquid-water sink and no subduction return: both halves of the loop are broken.

Initial Volatile Inventories



- Comparative accretion of Venus, Earth, Mars
- Initial water + carbon inventories overlap substantially
- Divergence is not from initial conditions alone
- Early evolution decides which branch each planet takes

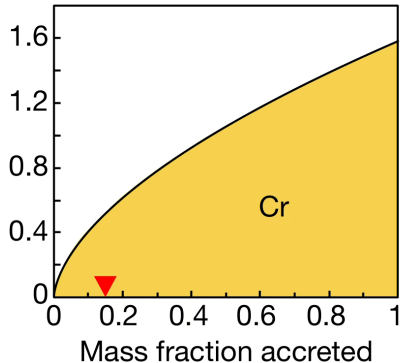
Credit: Lammer et al. (2018)

Why the Twins Diverged: Habitability Implications

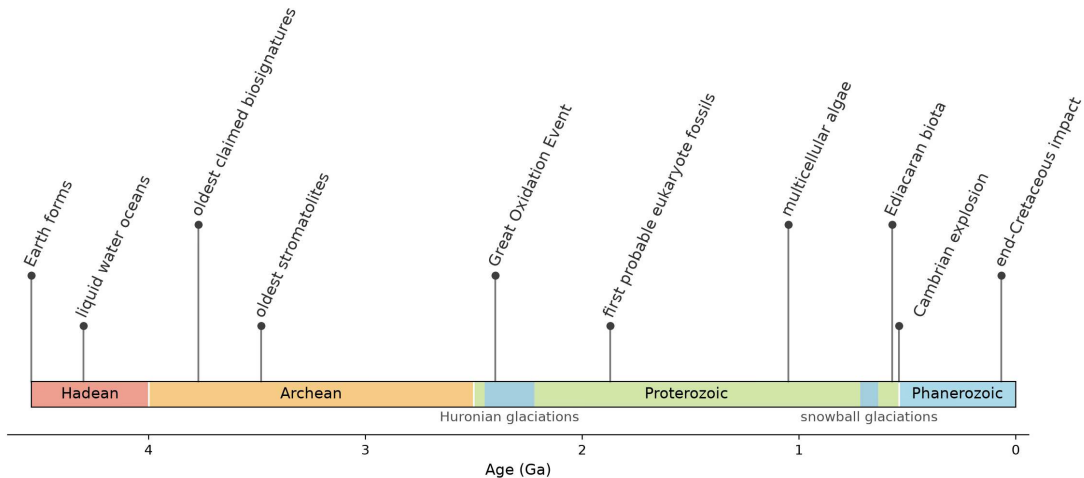
Coupled factors drove Earth and Venus apart:

1. **Solar flux:** $1.91\times$ Earth's insolation, above the runaway threshold
2. **Geophysics:** no intrinsic dynamo, slow retrograde rotation
3. **Tectonics:** a stagnant lid removes the subduction return leg of the carbon cycle

Venus defines the inner edge of the habitable zone: coupled interior-atmosphere feedbacks decide whether a terrestrial planet keeps its water.



Probability density for the chromium-bearing fraction of Earth's accreting mass against the cumulative accreted mass fraction: 95% of the mantle's Cr arrived during the final 85% of accretion, close to mass-proportional delivery, unlike Mo and Ru, which are back-loaded to the final stages. With the O, Ti, Ni, Mo and Ru tracers this reconstructs three accretion stages: the first 60% of Earth's mass about 51% enstatite-like, 40% ordinary-chondrite and 9% carbonaceous material, the rest essentially enstatite-like. The carbonaceous component, and with it most of Earth's water, came early rather than in the late veneer.



Major milestones in the history of life on Earth across 4.54 Gyr. Continuous habitability supported the transition from early microbial life to complex organisms, an evolutionary path denied to Venus. Credit: Course-original figure.



Modern stromatolites in the hypersaline water of Shark Bay, Western Australia: microbial mats trap and bind sediment into layered mounds. Fossil stromatolites of the same construction in the nearby Pilbara, dated to 3.48 Ga, are the oldest widely accepted evidence for life on Earth. Photo by Paul Harrison (Wikimedia Commons), [CC BY-SA 3.



8. Recent Advances and Future Missions



VERITAS, DAVINCI and EnVision over an Akatsuki UV view of Venus. Credit: Widemann et al. (2023)



The 2030s Venus fleet: DAVINCI (NASA), VERITAS (NASA), and EnVision (ESA). DAVINCI's descent probe mass spectrometer will measure atmospheric noble gases to distinguish early against late water loss. Widemann et al. (2023).

Summary

1. **Bulk twin:** same size and density, but ~ 500 K and 92 bar apart
2. **Runaway limit:** the Simpson-Nakajima threshold ($280\text{--}310 \text{ W m}^{-2}$) sets the inner habitable zone
3. **D/H evidence:** ~ 150× Earth's ratio records massive water loss
4. **Broken cycle:** no surface water stops the weathering sink; a stagnant lid removes the subduction return leg

Next: Lecture 10 Mercury and Mars.



Questions?

Next: Lecture 10. Rocky Planets II: Mercury & Mars

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience

